

Praveena Ganesan

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SUMMARY

Early-career software engineer (B.Tech, NIT Trichy, 2025) with 1 year of industry experience owning production systems end to end: real-time data pipelines, reconnect/recovery handling, and validation on model output before it can act. Heavy daily use of AI and agentic tools; strong debugging across multi-part systems.

EDUCATION

B.Tech, Electronics and Communication Engineering, National Institute of Technology, Tiruchirappalli

2021 – 2025

EXPERIENCE

AI Engineer (Agentic Systems), Sutherland Global Services

Oct 2025 – Present

Real-Time Multimodal AI Agent (Gemini Live + Robotics-ER) – [Demo](#)

- Worked with the robotics team on Gemini Robotics-ER 2 integration, converting natural-language commands and camera input into structured action plans for a robotic pick-and-place system, supporting 3 gripper types and 7 executor configurations.
- Implemented a real-time streaming pipeline using the Gemini Live API's bidirectional WebSocket session, moving from a turn-based flow toward continuous audio/video input and mid-task interruption; reduced live-session startup latency from ~1.48 s to ~0.66 s in local benchmarks.
- Designed and implemented a typed asyncio pub/sub event bus to connect perception, model responses, tool execution, and the UI, allowing the reasoning components to be tested independently from camera and robot hardware.
- Added checks around model-generated actions (depth validation, duplicate-target detection, JSON schema validation) and worked on reconnect/recovery handling so dropped live sessions could resume without restarting the entire task.

AI Engineering Platforms (AMS Buddy 2.5, CodeBuddy 2.5) – React.js, Python, FastAPI

- Supported a production AI incident co-pilot workflow for resolving customer incidents (signals, evidence, and timelines into probable causes, with actions gated by human approval) and a delivery-analytics platform automating reporting by integrating Azure DevOps, Jira, and SonarQube data, cutting manual effort 50%.

AI/ML Research Intern, NIT Trichy

Jan 2025 – May 2025

- Built a full-stack chatbot prototype (Flask, HTML/CSS, JavaScript) with REST inference endpoints for real-time NLU and MongoDB session logging.
- Trained and benchmarked 6 multi-intent classifiers (LR, SVM, RNN, Bi-LSTM, GRU, fine-tuned BERT) on ATIS and SNIPS with NLTK/Gensim preprocessing (TF-IDF, Word2Vec, Doc2Vec), reaching 92–94% accuracy.

Summer Research Intern, NIT Trichy

May 2024 – Jul 2024

- Built and deployed a low-latency image-classification REST API (FastAPI, TensorFlow 2.5 CNN) served asynchronously on Uvicorn (ASGI), improving inference 20% and handling 50+ concurrent requests.
- Improved CNN accuracy 15% with data augmentation, dropout, batch normalization, and early stopping; preprocessed 10,000+ images with Pillow, NumPy, and Matplotlib.
- Benchmarked SVM, GridSearchCV-tuned SVM, and CNN models; the CNN achieved 25% higher accuracy.

PROJECTS

MedFlow AI – Next.js, Node.js/Express, MongoDB, BullMQ, Redis, AWS S3, Docker

[GitHub](#)

- Built a healthcare platform with appointment scheduling, conflict prevention, a lightweight EHR, and role-based access.
- Built a BullMQ + Redis document-extraction pipeline (~8 s/20 pages), load-tested at 500+ concurrent requests (sub-200 ms p95).
- Added a RAG chatbot using embeddings and vector search (ChromaDB/Pinecone) with ~87% top-4 retrieval accuracy.

Open-Source Contributions

Ongoing

- [LiteLLM Gateway \(PR #38266\)](#): Diagnosed and fixed a parameter-leakage security bug in a microservices API gateway, closing a cascading regression via a targeted gating mechanism; backed by 195+ new tests (313/313 passing); PR in maintainer code review.
- [Mem0 \(Issue #6427\)](#): Root-caused and fixed a JSON-parsing bug (regex/fence-stripping conflict) in a memory system.

SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL

Frontend & Backend: React, Next.js, FastAPI, Flask, REST APIs, Node.js, Express, BullMQ, WebSockets, asyncio, PyTest

Databases & Cloud: MongoDB, Redis, AWS (S3), Docker, CI/CD, Git, Observability (logging, metrics, alerting)

AI/ML (applied): PyTorch, LLM Agents, LangGraph, MCP, Function Calling, Gemini API, RAG, Embeddings, Vector Databases (ChromaDB, Pinecone), TensorFlow; daily use of GitHub Copilot and Claude

Certifications: Azure Developer Associate (AZ-204), AWS Solutions Architect – Associate; Codeforces Rating 1600+